P-L-137

FEBRUARY 28, 1949

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

LAUNDRY SOUR; FLUORIDE TYPE

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than June 1, 1949, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A-1. The following Federal Specifications, of the issues in effect on the date of invitation for bids, shall form a part of this specification:

P-S-530—Soap and Soap-Products; General Specifications (Methods for Sampling and Testing).

RR-S-366—Sieves; Standard, Testing.

B. TYPE AND GRADE

B-1. Laundry sour in accordance with this specification shall be furnished in only one type and grade. (See par. I-3.)

C. MATERIAL

C-1. The laundry sour shall be a free flowing homogeneous mixture of sodium silicofluoride and sodium acid fluoride.

D. GENERAL REQUIREMENTS

D-1. See section E.

E. DETAIL REQUIREMENTS

E-1. *Form.*—The laundry sour shall be furnished in either powder or crystalline form. There shall be no indications of "caking" in the container when received.

E-2. *Composition.*—The laundry sour shall contain not less than 95 percent, by weight, of a mixture of sodium silicofluoride and sodium acid fluoride neither of which shall be less than 42.5 percent of the total composition. The laundry sour shall not contain wetting agents or impurities or adulterants which may be injurious to fabrics or laundry equipment.

E-3. *Oxalic acid.*—The laundry sour shall meet the test requirements of paragraph F-2b with respect to the oxalic acid content.

E-4. *Solubility.*—The insoluble matter shall not exceed 5 percent by weight when tested in accordance with the method described in paragraph F-2c.

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E-5. *Particle size.*—Not less than 85 percent, by weight, of the laundry sour shall pass through a No. 80 sieve (Fed. Spec. RR-S-366) when tested as described in paragraph F-2d.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F-1. *Sampling.*—When practicable, original unopened containers shall be taken at random from each delivery for test. When this is not practicable, a representative sample shall be taken, placed in clean, dry, glass containers, sealed and sent to the laboratory. The sample shall consist of not less than 1 pound nor more than 5 pounds.

F-2. *Inspection and tests.*—

F-2a. *Composition.*—

F-2a(1). *Sodium silicofluoride (Na_2SiF_6).*—A specimen of 0.5 ± 0.005 gram shall be accurately weighed and placed in a large platinum dish. Sufficient alcoholic potassium chloride and sodium carbonate solution (par. F-2a(1)a) shall be added rapidly with constant stirring to insure alkalinity. (The solution shall not be allowed to become acid.) The stirring shall be continued until all of the soluble material in the specimen has dissolved. The solution shall be filtered with suction and the residue shall be washed with alcoholic potassium chloride solution (par. F-2a(1)b) until the wash liquid is not alkaline to phenolphthalein. The filter and residue shall be transferred to a beaker and 100 milliliters of carbon-dioxide-free distilled water shall be added. The beaker and contents shall be heated to 70° to 90°C. and with heating continued shall be titrated in such a manner that the titration shall be completed with the solution actively boiling. Titration shall be made with an accurately standardized 0.2 normal sodium hydroxide solution (free from carbonates) using phenolphthalein as an indicator. The number of milliliters of standard sodium hydroxide solution used in the titration shall be designated as *B* and the normality of the sodium hydroxide solution as *N*. Then: Sodium silicofluoride, percent = $B \times N \times 0.403$.

NOTE.—Each milliliter of 1.0 *N* NaOH solution is equivalent to 0.047 gram of sodium silicofluoride.

F-2a(1)a. *Alcoholic potassium chloride and sodium carbonate solution.*—15.0 ± 0.1 grams of reagent-grade potassium chloride shall be dissolved in 100 milliliters of cool, carbon-dioxide-free distilled water; 100 milliliters of neutral 95-percent ethyl alcohol shall be added and the solution shall be well shaken. A sufficient quantity of reagent-grade sodium carbonate shall be added to form a saturated solution.

F-2a(1)b. *Alcoholic potassium chloride solution.*—45.0 ± 0.1 grams of reagent-grade potassium chloride shall be dissolved in 300 milliliters of cool, carbon-dioxide-free distilled water; 300 milliliters of neutral 95-percent ethyl alcohol shall be added to the solution and the whole shall be well shaken.

F-2a(2). *Sodium acid fluoride, ($NaHF_2$).*—A specimen of 0.5 ± 0.003 gram shall be accurately weighed and placed in a large platinum dish. Fifty milliliters of carbon-dioxide-free distilled water and a few drops of phenolphthalein indicator shall be added

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to the specimen. The resulting solution shall be titrated with 0.2 normal sodium hydroxide solution free of carbonates, and accurately standardized. When the fading of the color becomes sluggish, the dish and contents shall be heated while titrating and the titration shall be completed with the solution boiling. The number of milliliters of standard sodium hydroxide solution used in the titration shall be designated as A, the number of milliliters of sodium hydroxide acid solution used in the titration for sodium silicofluoride shall be designated as B, and the normality of the sodium hydroxide solution as N.

Sodium acid fluoride, percent $= (A - B) \times N \times 12.4$.

NOTE.—Each milliliter of 1.0 N NaOH solution is equivalent to 0.002 gram of sodium acid fluoride.

F-2b. *Oxalic acid.*—A specimen of 2.0±0.1 grams shall be dissolved in distilled water and the volume of the solution made up to 100 milliliters. Five milliliters of the solution shall be pipetted into a beaker containing litmus paper. The 5-milliliter portion shall be acidified with dilute sulfuric acid (10 percent) and the whole heated to 70° to 75°C. One milliliter of dilute potassium permanganate solution (50 mg. of potassium permanganate in 100 ml. of distilled water) shall be added. The color due to the permanganate solution shall not be destroyed within 3 minutes.

F-2c. *Solubility.*—Five hundred milliliters of distilled water shall be heated to a temperature of 87° to 88°C. in an Erlenmeyer flask. A 1.000±0.005-gram specimen of the sour shall be added to the water in the flask. The flask shall be stoppered and shaken vigorously for 30 seconds. The contents of the flask shall be rapidly filtered through a weighed Gooch crucible containing a suitable asbestos mat (or a sintered glass crucible) by decantation. Do not stir the solution. The entire filtration shall be completed in less than 2 minutes. Any solid material remaining in the flask shall be washed into the filter by three separate small portions (not exceeding 15 ml. per portion) of distilled water at a temperature of 20°±2°C. Do not give the crucible and contents any additional washing. The crucible and contents shall be dried to constant weight at 105° to 110°C.

$$\text{Insoluble matter, percent} = \frac{(\text{Weight of crucible + contents after drying}) - (\text{weight of crucible and pad})}{\text{Weight of specimen}} \times 100$$

F-2d. *Particle size.*—The particle size shall be determined as described in the applicable portions of method 210.1 of Federal Specification P-S-586.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G-1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable under this specification.

G-2. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate to the point of delivery called for in the contract or purchase order.

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G-3. *Marking.*—

G-3a. *Issue packages.*—Unless otherwise specified, issue packages shall be marked with the name of the material, the quantity contained therein, and the name of the manufacturer or contractor.

G-3b. *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the quantity contained therein as defined by the contract or order, under which the shipment is made, the name of the contractor and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H-1. The following departmental specifications, of the issues in effect on the date of invitation for bids, shall form a part of this specification and shall be applicable to purchases made under this specification by the respective departments.

H-2. *Army.*—

H-2a. *Applicable specifications.*—

Joint Army-Navy Specifications:

JAN-P-112—Packaging and Packing for Overseas Shipment—Drums, Plywood.

JAN-P-124—Packaging and Packing for Overseas Shipment—Containers (Cans, Pails, and Drums, Metal.

JAN-P-134—Packaging and Packing for Overseas Shipment—Kegs, Slack.

U. S. Army Specification:

100-2—Standard Specification for Marking Shipments by Contractors.

(Copies of specifications required for Army purchases may be obtained from the procuring agency. Both the title and identifying symbol number should be stipulated when requesting copies.)

H-2b. *Packing.*—

H-2b(1). *For domestic shipment.*—One hundred pounds of the product shall be packed for domestic shipment in any one of the following types of shipping containers:

(a) Plywood drum, type A or B, conforming to the requirements of Joint Army-Navy Specification JAN-P-112, except that water-resistant plywood, external toxic water-repellent treatment, cross battens, and strapping will not be required.

(b) Slack keg, conforming to the requirements of Joint Army-Navy Specification JAN-P-134, except that head cleats and strapping will not be required.

(c) Fiber drum, conforming to the requirements of Consolidated Freight Classification No. 17, Rule 41, Section 11. The sidewall of the drum shall be fabricated of multiple plies of kraft liner board with no ply less than 0.012 inch thick. The sidewall shall be either continuous without joint (convolutely wound) or with lap joint. Outer surface shall be waterproofed and all plies firmly glued together with good quality adhesive. The overlap, on the lap joint sidewall, shall be not less than 2 inches and shall be firmly glued together throughout the entire area of contact. The joint shall be reinforced by a double row of metal rivets or

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staples or by lock-stitch cotton thread in accordance with standard commercial practice. Headings shall be either of fiberboard having a minimum thickness of 0.120 inch with the outer surface waterproofed, of metal having a minimum thickness of 20 gage, of wood having a minium thickness of nominal 1-inch material, or of 3-ply plywood, having a minimum thickness of ¾ inch. Joints in wooden heads must be tight and firmly glued. Fiberboard sidewalls and headings shall have a bursting strength of not less than 600 pounds per square inch (Mullen or Cady). The drums shall be closed and sealed in accordance with any one of the following standard commercial methods: friction type covers, securely held in place by crimped, or bolted-type lugs, or by use of plug expanding tool, or lever-actuated locking bands, or slip-on covers secured by sealing tape. Filled and closed drums shall withstand the drop test outlined in Consolidated Freight Classification No. 17, without leakage of contents or separation of component parts of the body or head, sufficient to destroy its utility as a shipping container.

H-2b(1)a. Each slack keg or plywood drum shall be provided with an inner bag liner, made of 45-pound-basis-weight creped kraft paper having a minimum stretch of 15 percent in both directions. The liner shall be sealed or otherwise closed to prevent sifting of contents during shipment.

H-2b(2). *For overseas shipment.*—The product shall be packed in a leakproof 5-gallon-capacity, round, open-top metal container having a full-removable bolted-ring or lug-cover closure conforming to the requirements of Joint Army-Navy Specification JAN-P-124, type VI. The container shall be made of 24-gage bonderized steel or phosphate-treated blackplate throughout. The container shall have a swaged or rolled bead reinforcement near the top and shall be offset to permit stacking. Interior and exterior coatings for the container shall be applied to comply with the requirements of the container specification.

H-2c. *Marking.*—

H-2c(1). *For domestic shipment.*—Unless otherwise specified, legible commercial marking of the shipping container is acceptable, provided that the following information is included:

Stock number
Nomenclature
Quantity
Gross weight and cube
Contract or requisition number
Name and address of manufacturer

H-2c(2). *For overseas shipment.*—Unless otherwise specified, shipping containers shall be marked in accordance with the requirements of U. S. Army Specification 100-2. Marking on metal containers shall be lithographed or stenciled with a noncorrosive and weather-resistant black ink.

H-2c(2)a. *Special marking.*—In addition to the marking specified above, the following shall be either lithographed or legibly stenciled on the upper two-thirds of the face of the shipping container in letters not less than ½ inch in height:

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THIS MATERIAL IS TOXIC
IF TAKEN INTERNALLY.
IT SHOULD NOT BE STORED OR
USED WHERE FOOD IS
PREPARED OR KEPT.

H-3. *Navy.*—

H-3a. *Applicable specifications and other publications.*—

Joint Army-Navy Specifications.

JAN-P-110—Packaging and Packing for Overseas Shipment—Drums, Metal, 55-Gallon (for Other than Petroleum Products).

JAN-P-112—Packaging and Packing for Overseas Shipment—Drums, Plywood.

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN-P-140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner

Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

Interstate Commerce Commission Publication:

Regulations for Transportation of Explosives and Other Dangerous Articles by Freight.

(Copies of Federal Specifications and Joint Army-Navy specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. When requesting specifications, refer to both title and number (or symbol).

(Information as to the availability of Interstate Commerce Commission Regulations for the Transportation of Explosives and Other Dangerous Articles by Freight, may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

H-3b. *Packing.*—

H-3b(1). *For domestic shipment.*—Laundry sour shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery. When metal drums or plywood drums are used, they shall conform to the requirements of Joint Army-Navy Specification JAN-P-110 or JAN-P-112, respectively. The gross weight of drums shall not exceed approximately 150 pounds.

H-3b(2). *For overseas shipment.*—Laundry sour shall be packed in metal drums or plywood drums conforming to the requirements of Joint Army-Navy Specification JAN-P-110 or JAN-P-112, respectively. Plywood drums shall be lined with a sealed waterproof

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bag, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN-P-125. The seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN-P-140. The gross weight of drums shall not exceed approximately 150 pounds.

H-3b(3). Exceptional commodities or especially dangerous articles may require better drums than are specified herein. In every case the container must comply with the specifications prescribed in the Interstate Commerce Commission Regulations for the Transportation of Explosives and Other Dangerous Articles for the particular articles to which those specifications apply.

H-3c. *Marking.*—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with the requirements of the Navy Shipment Marking Handbook.

H-3d. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See pars. H-3b and H-3c.)

I. NOTES

I-1. Purchasers should exercise any desired options offered herein. (See pars. G-1, G-2, and G-3.)

I-2. The laundry sour covered by this specification is intended to be used principally by dusting the dry material on the revolving cylinders of washing machines.

I-3. If an acetic acid sour is desired, it may be prepared by diluting the acid covered by Federal Specification O-A-76. If an oxalic acid sour is desired, it may be procured under Federal Specification O-A-91.

I-4. It is believed that this specification adequately describes characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I-5. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

I-6. An Index of Federal Specifications may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

I-7. Copies of this specification and of P-S-536 and RR-S-366 may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. The price of this specification and of RR-S-366, 5 cents each; P-S-536, 10 cents.

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